CloudPulse

DevOps Monitoring and Alerting Sandbox | Project Report

1. Introduction

CloudPulse is a local DevOps monitoring and alerting platform built to monitor system resources, collect application metrics, visualize performance data, and manage alerts. It runs inside an Ubuntu 22.04 virtual machine provisioned with Vagrant and VirtualBox. Docker Compose runs the application and monitoring services in containers.

2. Abstract

The project combines infrastructure provisioning, containerization, metrics collection, visualization, and alert management in one local environment. Node Exporter exposes host metrics, Prometheus collects metrics and evaluates alert rules, Grafana displays monitoring data, and Alertmanager routes notifications to the Flask application webhook. HTTP error simulation supports alert testing.

3. Tools Used

- **Ubuntu 22.04 LTS:** Operating system for the virtual machine.
- **VirtualBox:** Runs the local virtual machine.
- **Vagrant:** Automates virtual machine provisioning.
- **Docker and Docker Compose:** Build and run application and monitoring containers.
- **Prometheus:** Collects metrics and evaluates eight predefined alert rules.
- **Grafana:** Visualizes system and application metrics.
- **Alertmanager:** Routes alerts and webhook notifications.
- **Node Exporter:** Exposes host CPU, memory, disk, and network metrics.
- **Python Flask:** Provides the web application, dashboard data, and alert webhook endpoint.

4. Steps Involved in Building the Project

1. **Environment setup:** Installed and configured VirtualBox and Vagrant.
2. **VM provisioning:** Used the Vagrantfile to create an Ubuntu 22.04 virtual machine with the required resources and network access.
3. **Container setup:** Prepared Dockerfile and Docker Compose configuration for Flask, Prometheus, Grafana, Alertmanager, and Node Exporter.
4. **Metrics collection:** Configured Prometheus scrape jobs for application and system metrics.
5. **Dashboard configuration:** Configured Grafana provisioning and dashboards to display monitoring data.
6. **Alert configuration:** Added eight Prometheus alert rules and configured Alertmanager to route alerts to the Flask webhook.
7. **Testing:** Checked containers and Prometheus targets, simulated HTTP 500 errors, and verified the HighHttpErrorRate alert and webhook delivery.

5. Conclusion

CloudPulse demonstrates an integrated local monitoring and alerting environment using DevOps tools. Verification confirmed five running Docker containers, four Prometheus targets in the UP state, eight loaded alert rules, an operational Grafana dashboard, and successful HTTP error alert delivery through Alertmanager to the Flask webhook. The project provided practical experience in VM provisioning, container orchestration, observability, alert configuration, and troubleshooting.